

# **Wazuh Configuration, Attack Simulation & Log Monitoring**

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Course: Certified IT Infrastructure & Cyber SOC Analyst (CICSA)



## **Acknowledgement**

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Athira Krishnan

## **Abstract**

This project explains how to install and configure a basic Security Operations Center (SOC) lab using Wazuh on a Linux system. The objective was to build a centralized log monitoring and threat detection setup, perform controlled attack simulations, and observe how security events are detected and displayed through a real-time dashboard.

The lab was set up in a virtual environment. I used Ubuntu to configure the Wazuh Manager and Agent, and Kali Linux as the attacker machine. After installing and configuring the required Wazuh components, I enabled log collection and rule-based detection. The dashboard interface allowed me to monitor events and alerts as they happened.

I carried out different attack simulations, including SSH brute-force attempts, reconnaissance, privilege escalation activities and file modification attacks. These actions produced real security logs, which were collected and analyzed by the Wazuh Manager. Based on predefined rules, alerts were triggered and categorized by severity levels, helping us clearly understand how suspicious activities are identified and prioritized.

Through this project, I gained practical exposure to SIEM deployment, event correlation, and continuous security monitoring. It helped me understand how centralized monitoring plays an important role in detecting unauthorized access, tracking authentication failures, and responding to potential cybersecurity threats in real-world environments.

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# 1. Introduction

A growing number of cybersecurity threats, including brute force attacks, malware infections, ransomware attacks, phishing and social engineering attacks, Distributed Denial-of-Service (DDoS) attacks, data breaches, zero-day exploits, web application attacks, and advanced persistent threats (APTs), are encountered by organizations and individuals in today's digital environment. Organizations rely on Security Information and Event Management (SIEM) systems and organized Security Operations Center (SOC) environments to efficiently monitor, identify, and address such threats. In order to reduce possible risks before they become serious security incidents, a SOC is essential for continuously evaluating system activity, connecting security events, and producing alerts.

The goal of this project is to leverage Wazuh, an open-source SIEM and host-based intrusion detection technology, to construct a simple SOC environment. The project's main goals are to set up Wazuh on a Linux-based system, simulate actual cyberattacks in a lab setting, gather and examine security logs, generate alerts based on detection rules, and visualize security events via a centralized dashboard.

A virtualized system including an Ubuntu server set up as the Wazuh Manager and another ubuntu as the Wazuh Agent and a Kali Linux computer used to conduct controlled attack simulations was used for the practical implementation. Through this project, key cybersecurity concepts such as log analysis, threat detection, rule-based alerting, and event correlation were explored.

This hands-on approach provides foundational knowledge of SOC operations and demonstrates the importance of continuous monitoring in maintaining system security and integrity.

## 2. Overview of Wazuh

Wazuh is a free and open-source security platform that unifies XDR and SIEM capabilities. It protects workloads across on-premises, virtualized, containerized, and cloud-based environments. Wazuh helps organizations and individuals to protect their data assets against security threats. It is widely used by thousands of organizations worldwide, from small businesses to large enterprises.

The Wazuh architecture components:

- **Wazuh Manager (Server):**  
The central component responsible for receiving logs from agents, analyzing events using predefined rules, correlating security data, and generating alerts.

- **Wazuh Agent:**  
Installed on monitored endpoints, the agent collects system logs, authentication logs, and security-related events, then securely forwards them to the Wazuh Manager for analysis.
- **Wazuh Indexer:**  
The indexer stores and indexes security event data, enabling efficient search, filtering, and retrieval of logs.
- **Wazuh Dashboard:**  
A web-based interface that allows users to visualize alerts, monitor security events, analyze trends, and manage agents in real time.

Together, these components create a centralized monitoring system that enables effective detection and analysis of cybersecurity threats within an organization.

### 3. Objectives

This project's main goal is to use Wazuh to create and implement a simple Security Operations Center (SOC) system in a controlled laboratory setting. The project's goal is to give participants hands-on experience with security monitoring, log management, and threat detection techniques by analyzing system activity in real time.

The specific objectives of this project are:

1. To install and configure Wazuh Manager and Agent on Linux-based systems.
2. To establish a centralized log collection and monitoring mechanism.
3. To simulate controlled cyberattacks.
4. To analyze security logs generated during attack simulations.
5. To generate alerts based on predefined detection rules.
6. To visualize security events and alert trends using the Wazuh Dashboard.
7. To understand the practical workflow of a Security Operations Center, including event correlation and incident monitoring.

## **4. Prerequisites**

### **3.1 Hardware Requirements**

- Minimum 8 GB RAM & 6 processors (Wazuh Manager)
- At least 50 - 70GB free disk space
- 2 - 4 GB RAM & 2 processors (Wazuh Agent and attacker Machine)

### **3.2 Software Requirements**

- Ubuntu 24.04 LTS – Installed as Wazuh Manager
- Ubuntu 22.04 LTS – Installed as Wazuh Agent
- Kali Linux – Used as attacker machine
- VirtualBox – Virtualization platform
- Wazuh 4.14
- Hydra – For SSH brute force simulation

### **3.3 Network and Configuration Requirements**

- NAT ( Internet access for updates and installations )
- Host-Only ( Internal secure communication between lab machines)

## **5. Procedures**

This section describes the technical steps followed to deploy and evaluate a Security Operations Center (SOC) environment using Wazuh. The implementation was carried out in a virtual lab setup, including installation of the Wazuh Manager and Agent, network configuration, attack simulation, log collection, alert analysis and dashboard creation. The structured approach ensured proper system integration and effective monitoring of security events.

### **5.1 Environment Preparation**

After installing VirtualBox as the virtualization platform, three virtual machines were created: Kali Linux (Attacker Machine), Ubuntu 24.04 LTS (Wazuh Manager), and Ubuntu 22.04 LTS (Wazuh Agent).

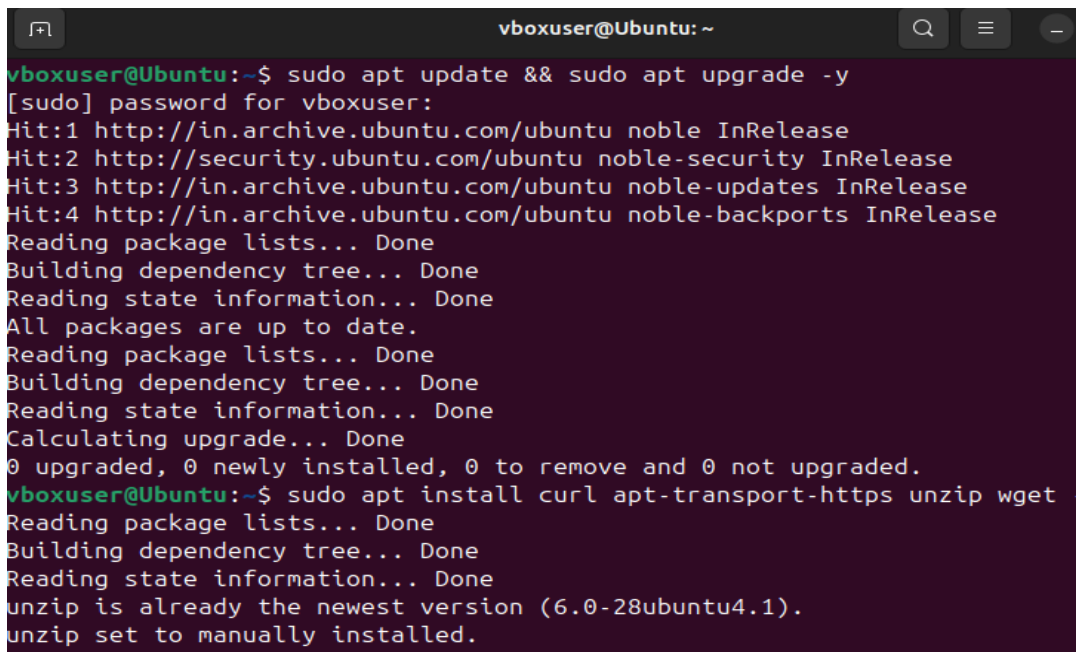
Set up two network adapters for virtual machines: a host-only adapter for internal communication

and a network address translation (NAT) for internet access. Ping was used to confirm connectivity between machines.

## 5.2 Installation of Wazuh Manager (Ubuntu 24.04)

Updated the system packages: `sudo apt update && sudo apt upgrade -y`.

Installed curl `sudo apt install curl apt-transport-https unzip wget -y` to fetch and download the official Wazuh installation script and required files directly from the Wazuh repository to the Ubuntu machine.

A terminal window titled 'vboxuser@Ubuntu: ~' with a dark background and light text. It shows the execution of two commands. The first command is 'sudo apt update && sudo apt upgrade -y', which prompts for a password and then shows the output of 'apt update' and 'apt upgrade'. The second command is 'sudo apt install curl apt-transport-https unzip wget -y', which shows the output of 'apt install' for these packages. The terminal output is as follows:

```
vboxuser@Ubuntu:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for vboxuser:
Hit:1 http://in.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
vboxuser@Ubuntu:~$ sudo apt install curl apt-transport-https unzip wget -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
unzip is already the newest version (6.0-28ubuntu4.1).
unzip set to manually installed.
```

### 5.2.1. Update system package & install curl

Installed Wazuh Manager (4.14), Wazuh Indexer, and Wazuh Dashboard on Ubuntu 24.04 LTS by downloading and running the official Wazuh installation script.

```
curl -s0 https://packages.wazuh.com/4.14/wazuh-install.sh && sudo bash ./wazuh-  
install.sh -a
```

This output shows the access credentials and a message that confirms that the installation was successful.



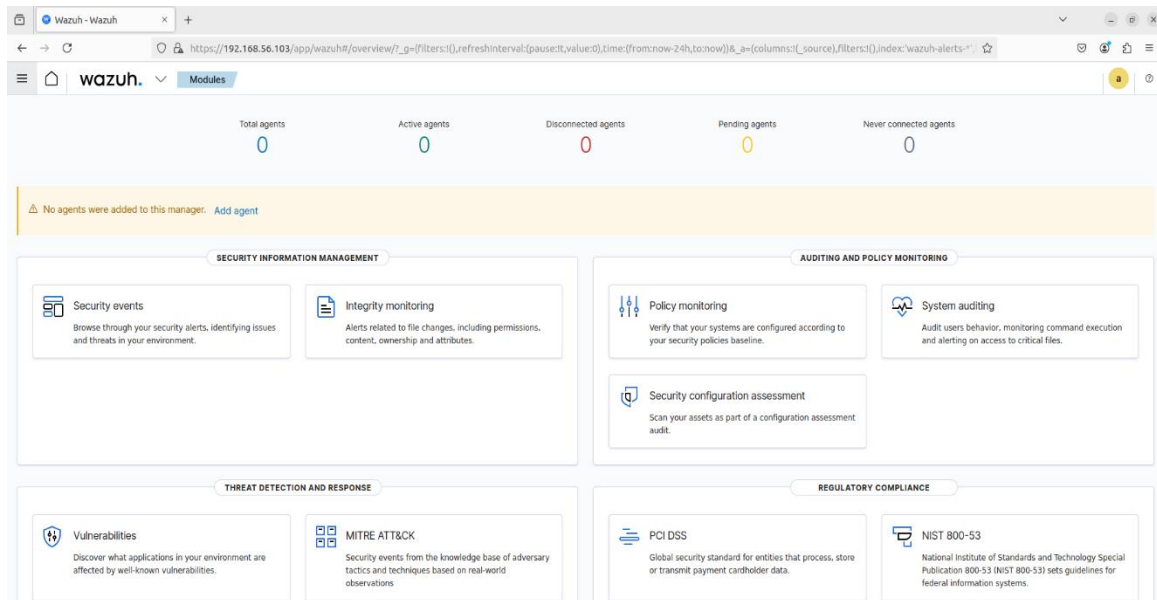
```
Firefox root@vbox-VirtualBox: /home/vbox
root@vbox-VirtualBox:/home/vbox# curl -sO https://packages.wazuh.com/4.14/wazuh-
install.sh && sudo bash ./wazuh-install.sh -a
1/03/2026 22:26:03 INFO: Starting Wazuh installation assistant. Wazuh version:
.14.3
1/03/2026 22:26:03 INFO: Verbose logging redirected to /var/log/wazuh-install.l
g
1/03/2026 22:26:09 INFO: --- Dependencies ---
1/03/2026 22:26:09 INFO: Installing gawk.
1/03/2026 22:26:16 INFO: Verifying that your system meets the recommended minim
m hardware requirements.
1/03/2026 22:26:16 INFO: Wazuh web interface port will be 443.
1/03/2026 22:26:23 INFO: --- Dependencies ---
1/03/2026 22:26:23 INFO: Installing debhelper.
1/03/2026 22:26:44 INFO: Wazuh repository added.
1/03/2026 22:26:44 INFO: --- Configuration files ---
1/03/2026 22:26:44 INFO: Generating configuration files.
1/03/2026 22:26:46 INFO: Generating the root certificate.
1/03/2026 22:26:46 INFO: Generating Admin certificates.
1/03/2026 22:26:47 INFO: Generating Wazuh indexer certificates.
1/03/2026 22:26:47 INFO: Generating Filebeat certificates.
1/03/2026 22:26:47 INFO: Generating Wazuh dashboard certificates.
1/03/2026 22:26:48 INFO: Created wazuh-install-files.tar. It contains the Wazuh
cluster key, certificates, and passwords necessary for installation.
1/03/2026 22:26:49 INFO: --- Wazuh indexer ---
```

### 5.2.2. Downloading & Installing Wazuh Manager

```
25/02/2026 17:38:35 INFO: Starting service wazuh-dashboard.
25/02/2026 17:38:36 INFO: wazuh-dashboard service started.
25/02/2026 17:38:56 INFO: Initializing Wazuh dashboard web application.
25/02/2026 17:38:56 INFO: Wazuh dashboard web application initialized.
25/02/2026 17:38:56 INFO: --- Summary ---
25/02/2026 17:38:56 INFO: You can access the web interface https://<wazuh-dashbo
ard-ip>:443
User: admin
Password: V4clEus56u1+3?wBjGkIX7Pbnqsc.n3B
25/02/2026 17:38:56 INFO: --- Dependencies ---
25/02/2026 17:38:56 INFO: Removing gawk.
25/02/2026 17:38:59 INFO: Installation finished.
athira@athira-VirtualBox:~$ sudo systemctl status wazuh-manager
● wazuh-manager.service - Wazuh manager
   Loaded: loaded (/lib/systemd/system/wazuh-manager.service; enabled; vendor
   Active: active (running) since Wed 2026-02-25 17:36:57 IST; 4min 49s ago
     Tasks: 161 (limit: 9429)
    Memory: 420.8M
       CPU: 43.172s
    CGroup: /system.slice/wazuh-manager.service
           └─48746 /var/ossec/framework/python/bin/python3 /var/ossec/api/scr▶
```

### 5.2.3. Credentials to access Wazuh Dashboard

Verified that all services were running properly and accessed the Wazuh dashboard via browser using the Wazuh manager IP address.



#### 5.2.4. Wazuh Dashboard

```
athira@athira-VirtualBox:~$ sudo systemctl status wazuh-manager
● wazuh-manager.service - Wazuh manager
   Loaded: loaded (/lib/systemd/system/wazuh-manager.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2026-02-25 17:36:57 IST; 4min 49s ago
     Tasks: 161 (limit: 9429)
    Memory: 420.8M
      CPU: 43.172s
   CGroup: /system.slice/wazuh-manager.service
           └─48746 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/manager.py
```

#### 5.2.5. Wazuh Manager status

```
vboxuser@Ubuntu: ~  
Feb 28 23:46:56 Ubuntu systemd[1]: Started wazuh-indexer.service - wazuh-indexer  
  
vboxuser@Ubuntu:~$ sudo systemctl status wazuh-dashboard  
● wazuh-dashboard.service - wazuh-dashboard  
   Loaded: loaded (/etc/systemd/system/wazuh-dashboard.service; enabled; pres  
   Active: active (running) since Sat 2026-02-28 23:46:07 IST; 4min 8s ago  
     Main PID: 803 (node)  
       Tasks: 11 (limit: 9428)  
      Memory: 337.0M (peak: 453.1M)  
         CPU: 21.018s  
    CGroup: /system.slice/wazuh-dashboard.service  
            └─803 /usr/share/wazuh-dashboard/node/bin/node --no-warnings --max  
  
Feb 28 23:47:00 Ubuntu opensearch-dashboards[803]: {"type":"log","@timestamp":>  
Feb 28 23:47:01 Ubuntu opensearch-dashboards[803]: {"type":"log","@timestamp":>  
Feb 28 23:47:01 Ubuntu opensearch-dashboards[803]: {"type":"log","@timestamp":>  
Feb 28 23:47:01 Ubuntu opensearch-dashboards[803]: {"type":"log","@timestamp":>  
Feb 28 23:47:01 Ubuntu opensearch-dashboards[803]: {"type":"log","@timestamp":>  
Feb 28 23:47:01 Ubuntu opensearch-dashboards[803]: {"type":"log","@timestamp":>  
Feb 28 23:47:01 Ubuntu opensearch-dashboards[803]: {"type":"log","@timestamp":>  
Feb 28 23:47:01 Ubuntu opensearch-dashboards[803]: {"type":"log","@timestamp":>  
Feb 28 23:47:01 Ubuntu opensearch-dashboards[803]: {"type":"log","@timestamp":>  
Feb 28 23:47:02 Ubuntu opensearch-dashboards[803]: {"type":"log","@timestamp":>  
lines 1-20/20 (END)
```

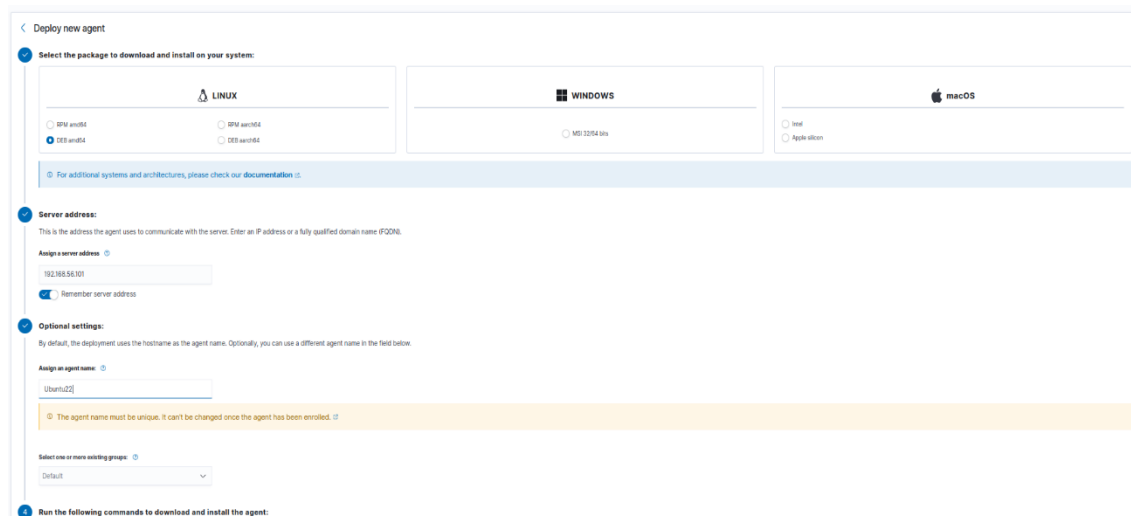
#### 5.2.5. Wazuh Dashboard Status

```
vboxuser@Ubuntu: ~  
vboxuser@Ubuntu:~$ sudo systemctl status wazuh-indexer  
● wazuh-indexer.service - wazuh-indexer  
   Loaded: loaded (/usr/lib/systemd/system/wazuh-indexer.service; enabled; pr  
   Active: active (running) since Sat 2026-02-28 23:46:56 IST; 1min 31s ago  
     Docs: https://documentation.wazuh.com  
     Main PID: 1459 (java)  
       Tasks: 105 (limit: 9428)  
      Memory: 1.6G (peak: 1.7G)  
         CPU: 1min 4.575s  
    CGroup: /system.slice/wazuh-indexer.service  
            └─1459 /usr/share/wazuh-indexer/jdk/bin/java -Xshare:auto -Dopense  
  
Feb 28 23:46:33 Ubuntu systemd-entrypoint[1459]: WARNING: System::setSecurityMa  
Feb 28 23:46:33 Ubuntu systemd-entrypoint[1459]: WARNING: Please consider repor  
Feb 28 23:46:33 Ubuntu systemd-entrypoint[1459]: WARNING: System::setSecurityMa  
Feb 28 23:46:35 Ubuntu systemd-entrypoint[1459]: Feb 28, 2026 6:16:35 PM sun.ut  
Feb 28 23:46:35 Ubuntu systemd-entrypoint[1459]: WARNING: COMPAT locale provide  
Feb 28 23:46:35 Ubuntu systemd-entrypoint[1459]: WARNING: A terminally deprecate  
Feb 28 23:46:35 Ubuntu systemd-entrypoint[1459]: WARNING: System::setSecurityMa  
Feb 28 23:46:35 Ubuntu systemd-entrypoint[1459]: WARNING: Please consider repor  
Feb 28 23:46:35 Ubuntu systemd-entrypoint[1459]: WARNING: System::setSecurityMa  
Feb 28 23:46:56 Ubuntu systemd[1]: Started wazuh-indexer.service - wazuh-indexe  
lines 1-21/21 (END)
```

#### 5.2.6. Wazuh-Indexer Status

## 5.3 Installation and Configuration of Wazuh Agent (Ubuntu 22.04)

A new agent was created through the Wazuh interface. After that, the Wazuh agent was installed and configured on Ubuntu 22.04. Lately, the agent was registered and connected to the Wazuh Manager by configuring it with the Manager's IP address. Verified Wazuh-agent connection in Wazuh dashboard.



### 4.3.1 New Wazuh Agent Creation

```
root@athira-VirtualBox: /home/athira
root@athira-VirtualBox: /home... x athira@athira-VirtualBox: ~ x athira@athira-VirtualBox: ~ x
athira@athira-VirtualBox:~$ sudo bash
root@athira-VirtualBox:/home/athira# wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent
wazuh-agent_4.14.3-1_amd64.deb && sudo WAZUH_MANAGER='192.168.56.101' WAZUH_AGENT_GROUP='default' W
AZUH_AGENT_NAME='Ubuntu22' dpkg -i ./wazuh-agent_4.14.3-1_amd64.deb
-2026-02-27 13:50:57- https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14
3-1_amd64.deb
resolving packages.wazuh.com (packages.wazuh.com)... 18.161.246.72, 18.161.246.120, 18.161.246.67, .
.
connecting to packages.wazuh.com (packages.wazuh.com)|18.161.246.72|:443... connected.
HTTP request sent, awaiting response... 200 OK
length: 13213254 (13M) [application/vnd.debian.binary-package]
saving to: 'wazuh-agent_4.14.3-1_amd64.deb.1'

wazuh-agent_4.14.3- 100%[=====] 12.60M 3.83MB/s in 3.3s

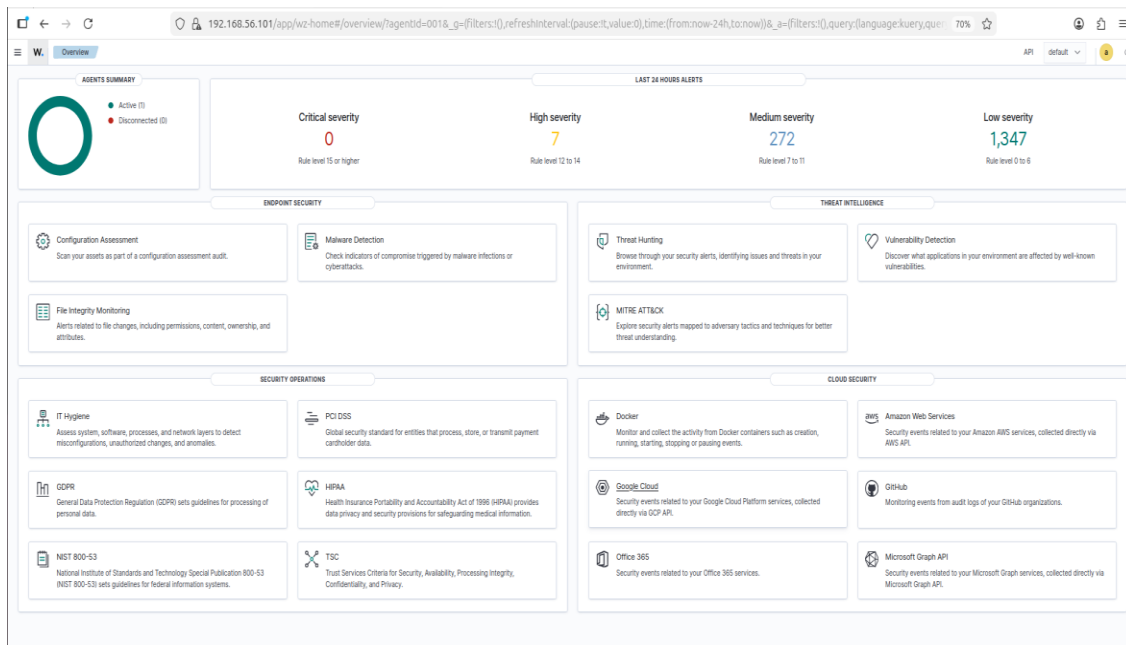
2026-02-27 13:51:00 (3.78 MB/s) - 'wazuh-agent_4.14.3-1_amd64.deb.1' saved [13213254/13213254]

(Reading database ... 288190 files and directories currently installed.)
Preparing to unpack .../wazuh-agent_4.14.3-1_amd64.deb ...
Unpacking wazuh-agent (4.14.3-1) over (4.14.3-1) ...
Setting up wazuh-agent (4.14.3-1) ...
```

### 4.3.2 Downloading & Installing Wazuh Agent

```
athira@athira-VirtualBox: ~  
root@athira-VirtualBox: /home... x athira@athira-VirtualBox: ~ x athira@athira-VirtualBox: ~ x  
athira@athira-VirtualBox:~$ sudo systemctl restart wazuh-agent  
sudo systemctl status wazuh-agent  
● wazuh-agent.service - Wazuh agent  
   Loaded: loaded (/lib/systemd/system/wazuh-agent.service; enabled; vendor p  
   Active: active (running) since Fri 2026-02-27 13:56:42 IST; 22ms ago  
   Process: 3678 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (co  
   Tasks: 31 (limit: 4599)  
   Memory: 34.9M  
   CPU: 3.626s  
   CGroup: /system.slice/wazuh-agent.service  
           └─3700 /var/ossec/bin/wazuh-execd  
             └─3711 /var/ossec/bin/wazuh-agentd  
               └─3724 /var/ossec/bin/wazuh-syscheckd  
                 └─3737 /var/ossec/bin/wazuh-logcollector  
                   └─3754 /var/ossec/bin/wazuh-modulesd  
  
Feb 27 13:56:34 athira-VirtualBox systemd[1]: Starting Wazuh agent...  
Feb 27 13:56:35 athira-VirtualBox env[3678]: Starting Wazuh v4.14.3...  
Feb 27 13:56:36 athira-VirtualBox env[3678]: Started wazuh-execd...  
Feb 27 13:56:37 athira-VirtualBox env[3678]: Started wazuh-agentd...  
Feb 27 13:56:38 athira-VirtualBox env[3678]: Started wazuh-syscheckd...  
Feb 27 13:56:39 athira-VirtualBox env[3678]: Started wazuh-logcollector...  
Feb 27 13:56:40 athira-VirtualBox env[3678]: Started wazuh-modulesd...  
Feb 27 13:56:42 athira-VirtualBox env[3678]: Completed.
```

#### 4.3.3. Wazuh-Agent Status



#### 4.3.4. Wazuh Agent in the Wazuh Dashboard.

And also ensured effective security event collection and analysis. Authentication logs from /var/log/auth.log were verified, and system logs including SSH and sudo activities were confirmed



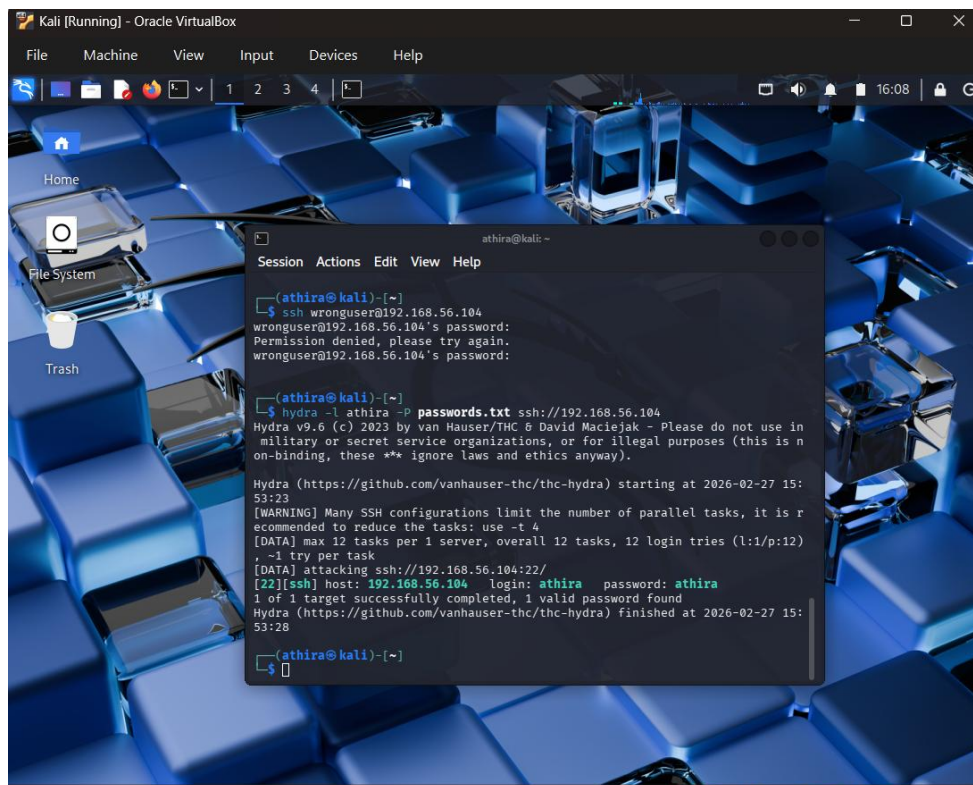
to be properly recorded and forwarded to the manager. Real-time event visibility was validated through the Wazuh Dashboard.

## 5.4 Attack Simulation

Attack simulations were conducted from Kali Linux to generate security events and tried 4 different types of attacks:

### SSH Brute Force Attack

- Created a password list file “password.txt” and used Hydra to attempt multiple failed SSH logins from Kali Linux to Ubuntu 22.04 Machine where agent was installed.



```
Kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
Home
File System
Trash
athira@kali ~
Session Actions Edit View Help
(athira@kali)~
$ ssh wronguser@192.168.56.104
wronguser@192.168.56.104's password:
Permission denied, please try again.
wronguser@192.168.56.104's password:
(athira@kali)~
$ hydra -l athira -P passwords.txt ssh://192.168.56.104
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in
military or secret service organizations, or for illegal purposes (this is n
on-binding, these ** ignore laws and ethics anyway).
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-02-27 15:
53:23
[WARNING] Many SSH configurations limit the number of parallel tasks, it is r
ecommended to reduce the tasks: use -t 4
[DATA] max 12 tasks per 1 server, overall 12 tasks, 12 login tries (l1:p:12)
, -1 try per task
[DATA] attacking ssh://192.168.56.104:22/
[22][ssh] host: 192.168.56.104 login: athira password: athira
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-02-27 15:
53:28
(athira@kali)~
$
```

#### 5.4.1. Brute Force Attack

### File Integrity / Modification Attack

- Modified /etc/passwd file of Ubuntu 22.04 Machine and generated the alert in Wazuh Dashboard.



#### 5.4.2. File Modification

### Port Scanning / Reconnaissance Attack

- Performed an Nmap port scan against the Ubuntu 22.04 machine and successfully generated reconnaissance alerts in the Wazuh Dashboard.

```

(kali㉿kali)-[~]
$ nmap -sS -sV -T4 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-03 15:45 IST
Stats: 0:00:13 elapsed; 0 hosts completed (0 up), 1 undergoing ARP Ping Scan
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Nmap scan report for 192.168.56.101
Host is up (0.0011s latency).
All 1000 scanned ports on 192.168.56.101 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:46:69:F2 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 13.74 seconds

(kali㉿kali)-[~]
$

```

#### 5.4.3. Port Scanning Attack

## Privilege Escalation Attempt

- Simulated a privilege escalation attempt on the Ubuntu 22.04 machine and successfully triggered authentication and sudo-related alerts in the Wazuh Dashboard.

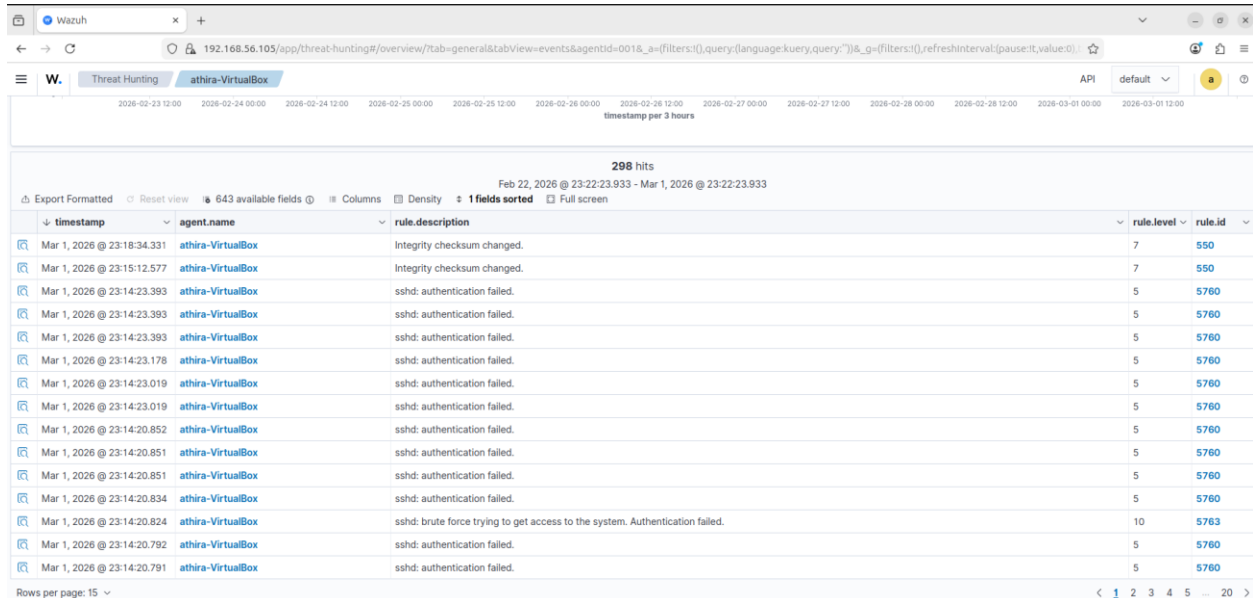
```
(kali㉿kali)-[~]  
$ ssh vboxuser@192.168.56.101  
vboxuser@192.168.56.101's password:  
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-14-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
Expanded Security Maintenance for Applications is not enabled.  
  
4 updates can be applied immediately.  
1 of these updates is a standard security update.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
Last login: Tue Mar  3 10:26:56 2026 from 192.168.56.103  
vboxuser@ubuntu:~$ sudo su  
[sudo] password for vboxuser:  
Sorry, try again.  
[sudo] password for vboxuser:  
Sorry, try again.  
[sudo] password for vboxuser:  
sudo: 3 incorrect password attempts  
vboxuser@ubuntu:~$ sudo -l  
[sudo] password for vboxuser:  
Sorry, try again.  
[sudo] password for vboxuser:  
Sorry, try again.  
[sudo] password for vboxuser:  
sudo: 3 incorrect password attempts  
vboxuser@ubuntu:~$ su root  
Password:  
su: Authentication failure  
vboxuser@ubuntu:~$
```

### 5.4.4. Privilege Escalation Attempt



## 5.5 Log Collection and Alert Generation

In this phase Wazuh agent forwarded logs to the manager and manager analyzed logs using predefined rules. Therefore, alerts were triggered based on suspicious behavior.



The screenshot shows the Wazuh Threat Hunting interface. The top navigation bar includes 'Wazuh', 'Threat Hunting', and 'athira-VirtualBox'. The main area displays a table of alerts for the agent 'athira-VirtualBox'. The table has columns for 'timestamp', 'agent\_name', 'rule.description', 'rule.level', and 'rule.id'. The alerts are sorted by timestamp, showing a series of 'sshd: authentication failed.' and 'Integrity checksum changed.' events. The interface also includes a search bar, a filter dropdown, and a 'Rows per page' selector.

| timestamp                  | agent_name        | rule.description                                                             | rule.level | rule.id |
|----------------------------|-------------------|------------------------------------------------------------------------------|------------|---------|
| Mar 1, 2026 @ 23:18:34.331 | athira-VirtualBox | Integrity checksum changed.                                                  | 7          | 550     |
| Mar 1, 2026 @ 23:15:12.577 | athira-VirtualBox | Integrity checksum changed.                                                  | 7          | 550     |
| Mar 1, 2026 @ 23:14:23.393 | athira-VirtualBox | sshd: authentication failed.                                                 | 5          | 5760    |
| Mar 1, 2026 @ 23:14:23.393 | athira-VirtualBox | sshd: authentication failed.                                                 | 5          | 5760    |
| Mar 1, 2026 @ 23:14:23.393 | athira-VirtualBox | sshd: authentication failed.                                                 | 5          | 5760    |
| Mar 1, 2026 @ 23:14:23.178 | athira-VirtualBox | sshd: authentication failed.                                                 | 5          | 5760    |
| Mar 1, 2026 @ 23:14:23.019 | athira-VirtualBox | sshd: authentication failed.                                                 | 5          | 5760    |
| Mar 1, 2026 @ 23:14:23.019 | athira-VirtualBox | sshd: authentication failed.                                                 | 5          | 5760    |
| Mar 1, 2026 @ 23:14:20.852 | athira-VirtualBox | sshd: authentication failed.                                                 | 5          | 5760    |
| Mar 1, 2026 @ 23:14:20.851 | athira-VirtualBox | sshd: authentication failed.                                                 | 5          | 5760    |
| Mar 1, 2026 @ 23:14:20.851 | athira-VirtualBox | sshd: authentication failed.                                                 | 5          | 5760    |
| Mar 1, 2026 @ 23:14:20.834 | athira-VirtualBox | sshd: authentication failed.                                                 | 5          | 5760    |
| Mar 1, 2026 @ 23:14:20.824 | athira-VirtualBox | sshd: brute force trying to get access to the system. Authentication failed. | 10         | 5763    |
| Mar 1, 2026 @ 23:14:20.792 | athira-VirtualBox | sshd: authentication failed.                                                 | 5          | 5760    |
| Mar 1, 2026 @ 23:14:20.791 | athira-VirtualBox | sshd: authentication failed.                                                 | 5          | 5760    |

### 5.5.1. Brute Force Alerts



### 5.5.2 Raw Events of Brute Force Attempt

## Document Details

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|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| manager.name            | vbox-VirtualBox                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| predecoder.hostname     | athira-VirtualBox                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| predecoder.program_name | sshd                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| predecoder.timestamp    | Mar 01 17:44:19                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| previous_output         | Mar 1 23:14:19 athira-VirtualBox sshd[5844]: Failed password for athira from 192.168.56.102 port 43892 ssh2<br>Mar 01 17:44:19 athira-VirtualBox sshd[5845]: Failed password for athira from 192.168.56.102 port 43896 ssh2<br>Mar 1 23:14:19 athira-VirtualBox sshd[5843]: Failed password for athira from 192.168.56.102 port 43880 ssh2<br>Mar 1 23:14:19 athira-VirtualBox sshd[5841]: Failed password for athira from 192.168.56.102 port 43884 ssh2 |
| rule.description        | sshd: brute force trying to get access to the system. Authentication failed.                                                                                                                                                                                                                                                                                                                                                                              |
| rule.firedtimes         | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| rule.frequency          | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| rule.gdpr               | IV_35.7.d, IV_32.2                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| rule.groups             | syslog, sshd, authentication_failures                                                                                                                                                                                                                                                                                                                                                                                                                     |
| rule.hipaa              | 164.312.b                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| rule.id                 | 5763                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| rule.level              | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| rule.mail               | false                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| rule.mitre.id           | T1110                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| rule.mitre.tactic       | Credential Access                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| rule.mitre.technique    | Brute Force                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| rule.nist_800_53        | SI.4, AU.14, AC.7                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| rule.nvds               | 11.4, 10.2.4, 10.2.5                                                                                                                                                                                                                                                                                                                                                                                                                                      |

### 5.5.3 Details of Brute Force Alert

## Document Details

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| Table                | JSON                                                                                                                                                                                                                                                                                                                |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| _index               | wazuh-alerts-4.x-2026.02.27                                                                                                                                                                                                                                                                                         |
| agent.id             | 001                                                                                                                                                                                                                                                                                                                 |
| agent.ip             | 192.168.56.104                                                                                                                                                                                                                                                                                                      |
| agent.name           | athira-VirtualBox                                                                                                                                                                                                                                                                                                   |
| decoder.name         | syscheck_integrity_changed                                                                                                                                                                                                                                                                                          |
| full_log             | File '/etc/passwd' modified<br>Mode: scheduled<br>Changed attributes: size,mtime,md5,sha1,sha256<br>Size changed from '3130' to '3124'<br>Old modification time was: '1772192352', now it is '1772192767'<br>Old md5sum was: '23ead6f3baec3393a099e4e5721fb26'<br>New md5sum is: '84f6ad11ch770d4c1da4dc1a8a280a1e' |
| id                   | 1772192802.2703805                                                                                                                                                                                                                                                                                                  |
| input.type           | log                                                                                                                                                                                                                                                                                                                 |
| location             | syscheck                                                                                                                                                                                                                                                                                                            |
| manager.name         | Ubuntu                                                                                                                                                                                                                                                                                                              |
| rule.description     | Integrity checksum changed.                                                                                                                                                                                                                                                                                         |
| rule.firedtimes      | 1                                                                                                                                                                                                                                                                                                                   |
| rule.gdpr            | II_5.1.f                                                                                                                                                                                                                                                                                                            |
| rule.gpg13           | 4.11                                                                                                                                                                                                                                                                                                                |
| rule.groups          | ossec, syscheck, syscheck_entry_modified, syscheck_file                                                                                                                                                                                                                                                             |
| rule.hipaa           | 164.312.c.1, 164.312.c.2                                                                                                                                                                                                                                                                                            |
| rule.id              | 550                                                                                                                                                                                                                                                                                                                 |
| rule.level           | 7                                                                                                                                                                                                                                                                                                                   |
| rule.mail            | false                                                                                                                                                                                                                                                                                                               |
| rule.mitre.id        | T1505.001                                                                                                                                                                                                                                                                                                           |
| rule.mitre.tactic    | Impact                                                                                                                                                                                                                                                                                                              |
| rule.mitre.technique | Stored Data Manipulation                                                                                                                                                                                                                                                                                            |
| rule.nist_800_53     | SI.7                                                                                                                                                                                                                                                                                                                |

### 5.5.4. Integrity Checksum changed Alert

Wazuh Threat Hunting interface showing a table of alerts. The table has columns: timestamp, agent.name, rule.descri..., rule.level, and rule.id. The alerts are for rule 52002, which is 'Apparmor DE...'. The agent.name is 'Ubuntu'. The timestamps range from Mar 3, 2026 @ 15:42:44.996 to Mar 3, 2026 @ 15:45:07.843. There is one alert for rule 533, 'Listened ports...', with a level of 7.

| timestamp                  | agent.name | rule.descri...    | rule.level | rule.id |
|----------------------------|------------|-------------------|------------|---------|
| Mar 3, 2026 @ 15:45:07.843 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:45:03.829 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:44:57.826 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:44:53.804 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:44:47.792 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:44:41.780 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:44:37.760 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:44:31.741 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:44:27.732 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:44:22.302 | Ubuntu     | Listened ports... | 7          | 533     |
| Mar 3, 2026 @ 15:44:21.708 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:43:51.468 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:43:47.459 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:43:41.398 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:43:27.296 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:43:21.183 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:43:17.132 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:43:11.115 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:43:05.095 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:43:01.014 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:42:55.004 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:42:50.998 | Ubuntu     | Apparmor DE...    | 3          | 52002   |
| Mar 3, 2026 @ 15:42:44.996 | Ubuntu     | Apparmor DE...    | 3          | 52002   |

### 5.5.5. Reconnaissance Alerts

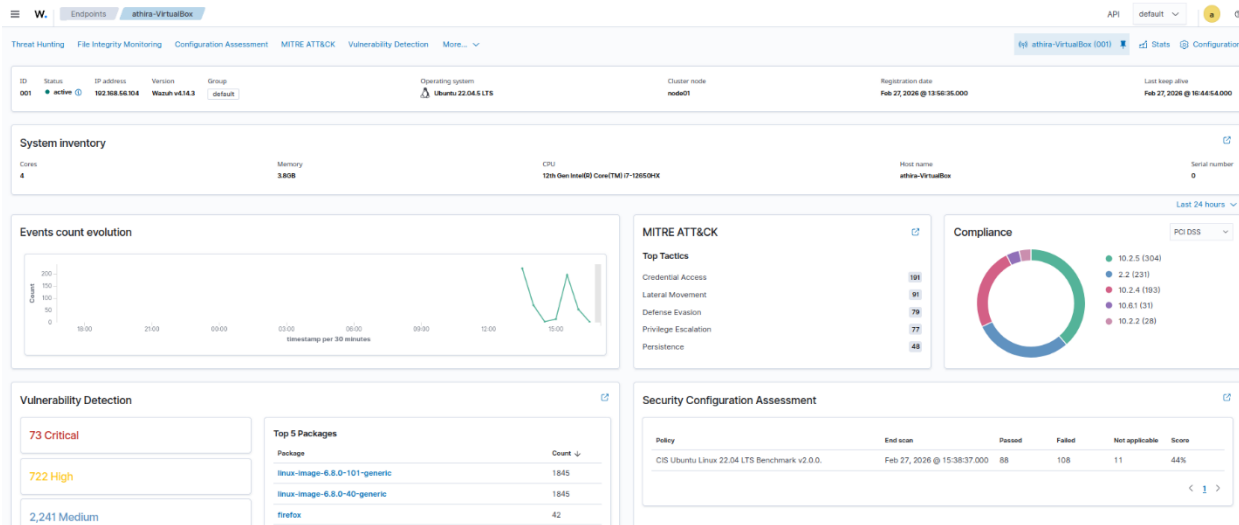
Wazuh Threat Hunting interface showing a table of alerts. The table has columns: timestamp, agent.name, rule.descri..., rule.level, and rule.id. The alerts are for rule 5301, 'User missed t...', and rule 5503, 'PAM: User logi...'. The agent.name is 'UbuntuServer24.04'. The timestamps range from Mar 3, 2026 @ 15:55:42.539 to Mar 3, 2026 @ 15:58:58.689.

| timestamp                  | agent.name        | rule.descri...     | rule.level | rule.id |
|----------------------------|-------------------|--------------------|------------|---------|
| Mar 3, 2026 @ 15:58:58.689 | UbuntuServer24.04 | User missed t...   | 5          | 5301    |
| Mar 3, 2026 @ 15:58:56.704 | UbuntuServer24.04 | PAM: User logi...  | 5          | 5503    |
| Mar 3, 2026 @ 15:58:48.692 | UbuntuServer24.04 | Three failed at... | 10         | 5404    |
| Mar 3, 2026 @ 15:58:38.707 | UbuntuServer24.04 | PAM: User logi...  | 5          | 5503    |
| Mar 3, 2026 @ 15:58:08.687 | UbuntuServer24.04 | Three failed at... | 10         | 5404    |
| Mar 3, 2026 @ 15:57:56.683 | UbuntuServer24.04 | PAM: User logi...  | 5          | 5503    |
| Mar 3, 2026 @ 15:57:14.710 | UbuntuServer24.04 | PAM: Login se...   | 3          | 5501    |
| Mar 3, 2026 @ 15:57:14.651 | UbuntuServer24.04 | sshd: authenti...  | 3          | 5715    |
| Mar 3, 2026 @ 15:57:04.654 | UbuntuServer24.04 | PAM: Login se...   | 3          | 5502    |
| Mar 3, 2026 @ 15:56:58.675 | UbuntuServer24.04 | PAM: Login se...   | 3          | 5501    |
| Mar 3, 2026 @ 15:56:58.622 | UbuntuServer24.04 | sshd: authenti...  | 3          | 5715    |
| Mar 3, 2026 @ 15:56:36.591 | UbuntuServer24.04 | sshd: Attempt ...  | 5          | 5710    |
| Mar 3, 2026 @ 15:56:32.601 | UbuntuServer24.04 | PAM: User logi...  | 5          | 5503    |
| Mar 3, 2026 @ 15:56:30.582 | UbuntuServer24.04 | sshd: Attempt ...  | 5          | 5710    |
| Mar 3, 2026 @ 15:56:14.584 | UbuntuServer24.04 | PAM: Login se...   | 3          | 5502    |
| Mar 3, 2026 @ 15:56:14.576 | UbuntuServer24.04 | PAM: Login se...   | 3          | 5501    |
| Mar 3, 2026 @ 15:56:14.575 | UbuntuServer24.04 | Successful su...   | 3          | 5402    |
| Mar 3, 2026 @ 15:56:14.575 | UbuntuServer24.04 | PAM: Login se...   | 3          | 5502    |
| Mar 3, 2026 @ 15:56:12.619 | UbuntuServer24.04 | PAM: Login se...   | 3          | 5501    |
| Mar 3, 2026 @ 15:56:12.565 | UbuntuServer24.04 | Successful su...   | 3          | 5402    |
| Mar 3, 2026 @ 15:55:42.541 | UbuntuServer24.04 | PAM: Login se...   | 3          | 5502    |
| Mar 3, 2026 @ 15:55:42.540 | UbuntuServer24.04 | PAM: Login se...   | 3          | 5501    |
| Mar 3, 2026 @ 15:55:42.539 | UbuntuServer24.04 | Successful su...   | 3          | 5402    |

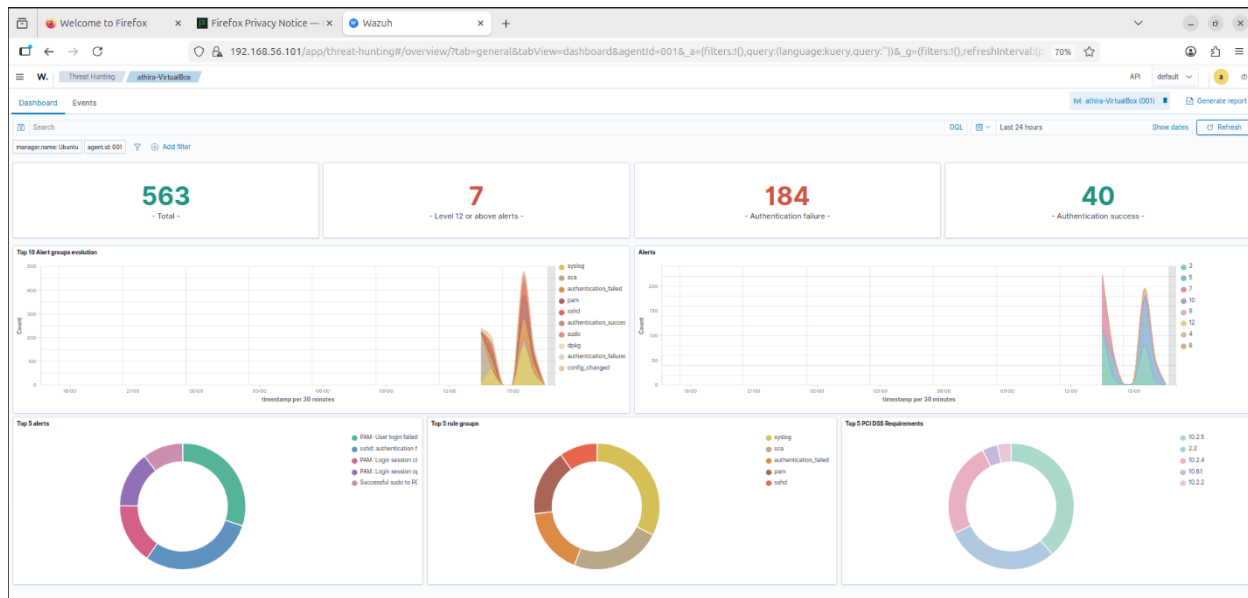
### 5.5.6. Privilege Escalation Alerts

## 5.6 Dashboard Analysis and Visualization

The status of the connected machine and the security alerts it has generated are displayed on the Wazuh Agent default dashboard. It shows basic information such as the operating system, IP address, and system name. It helps monitor activities and detect suspicious actions on the endpoint machine.

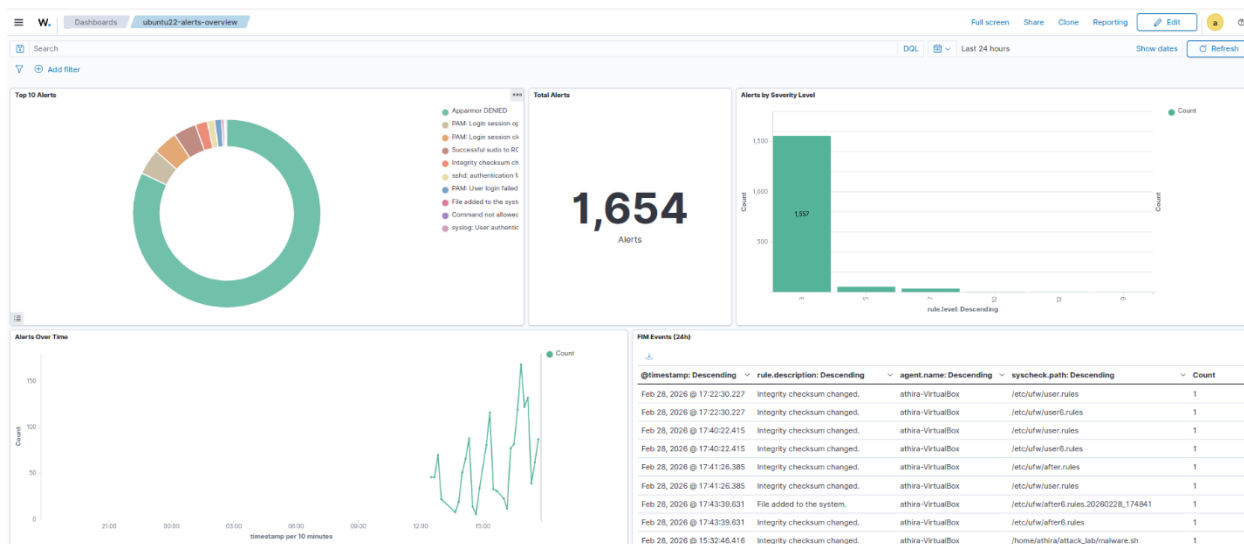


### 5.6.1. Agent-Dashboard



### 5.6.2. Agent-Thread Hunting Dashboard

A customized dashboard was created based on the last 24 hours of data, including the Top 10 alerts, total alert count, alerts by severity level, file integrity monitoring (FIM) events, and alerts over a time period. The dashboard was also saved under the name 'ubuntu24\_alerts\_overview' for future reference.



5.6.3. Customized Dashboard for Ubuntu 22 Machine

## 6. Challenges

- One of the challenges in the lab setup was ensuring compatibility between Ubuntu OS version and Wazuh server version.
- Wazuh components require high memory and processing power, especially in a virtual machine environment, which leads to slow performance and occasional service issues.

## 7. Conclusion

This project demonstrated the setup and implementation of a basic security monitoring lab environment. In a controlled virtual environment, a centralized monitoring system was put up to gather logs, identify suspicious activity, and produce security alerts.

The project provided hands-on experience in log analysis, alert generation, and security event monitoring. During the setup procedure, a number of technical difficulties were encountered, such as system compatibility issues and configuration concerns. Addressing these issues helped improve troubleshooting abilities and understanding of how monitoring systems operate in real-world environments.

The project highlighted the importance of continuous monitoring, proper configuration, and timely detection of potential security incidents. All things considered, this lab experience enhanced fundamental understanding of security monitoring and incident detection while developing useful abilities relevant to entry-level cybersecurity and SOC positions.

## 8. References

1. <https://documentation.wazuh.com/current/getting-started/index.html>
2. <https://youtu.be/cH4psUe-7P4?si=ZIPZUCahYYPYdWLv>
3. <https://www.youtube.com/watch?v=awOZ89VF56Y&t=3s>
4. <https://www.youtube.com/watch?v=QrcAhd5P7xw>
5. <https://www.geeksforgeeks.org/ethical-hacking/introduction-to-wazuh>